

Examiner reports

Q1.

In part (a), most correctly read the graph and then correctly used the coordinates to give the required P and V values.

The majority seemed well practised in the first step for part (b), writing the equation as $\log_{10}P = \log_{10}C + d\log_{10}V$. Many progressed to a correct solution easily and completed the question. Others became confused about what was a log and what was not or failed to handle the negative values and negative gradient. Those who tried to use point A and another point to generate simultaneous equations usually became lost at some point in the algebra.

In part (c), most remembered to include the units, relating the mathematical model back to the physical situation behind it.

Q2.

This question required the use of the exponential function and natural logs which produced a diverse range of responses. In part (a) about a third of students were able to correctly write down the required two equations using the information given in the question, but solving them simultaneously proved beyond many students. Some used the given value of b to find the value of a , which gained 1 mark and it also enabled them to tackle later parts of the question.

There were many correct responses to part (b) which was well done as long as students had found the correct values of a and b .

In part (c) students mostly failed to use inequalities but this approach did not lose marks. Those students who isolated the exponential term before taking logs were usually more successful than those who took logs straight away, as errors occurred later when simplifying the log terms. Many lost the final mark because they did not quote a year as their final answer as required. Appropriate numerical approaches were accepted and there was some follow through of the students value of ' a ', provided $b = 90$ was used.

There were some very good responses to part (d) and over half of all students were able to give a limitation of the model. It was disappointing that over 17% of students did not attempt this question despite it not being dependent upon any of the earlier work in the question.

Q3.

Part (a) was completed successfully by the majority of students. It was pleasing to see students being given relatively little structure and demonstrating that they could piece together the given information to solve the problem.

In part (b), around a third of students found difficulty in translating the problem given in context into the correct mathematical process. The most common mistake was not realising that the mass had to reach 280 mg before the next cup of coffee could be drunk. Some students failed to answer the question being asked and left their answers as 2 hours 56 minutes. This was another example where efficient use of a calculator could have helped some students.

Part (c) was testing the ability of students to criticise the model used in this context. Many students gave one word answers or incomplete sentences. While essays are not required in these types of question, and students should be encouraged to be concise, they do need to make themselves clear as examiners are not permitted to guess what they mean.

Q7.

The range of marks seen in this question typically varied between up to 4, scored by about 85% of the students, to up to 7 scored by about 25%. Very few completed the whole question successfully, with many making only superficial attempts at part (b).

Part (a)(i) and (ii). Most students answered correctly with only the occasional arithmetic error.

Part (a)(iii). Many students scored 2 out of the 3 marks available; although they understood what to do and followed correct manipulation, taking logarithms correctly, they didn't respond to the request for integers in the final expression. Those students who didn't simplify the expressions in their working often made an error.

Part (b)(i). Students were expected to attempt the required differentiation using the quotient or chain rule. Those that did this often made a sign error or had a wrong coefficient, sometimes including a t in their expression. Although some excellent algebra was seen in deriving the given result using the initial expression for N , many were unsuccessful in eliminating the $9e^{\frac{1}{3}}$ term, and some gave up without trying. However, many students tried to rearrange the given equation for N to make $e^{\frac{1}{3}}$ or t the subject, and then differentiate. It is difficult to see why they considered this approach would be helpful, but many made an algebraic error and little progress. Some tried to solve the differential equation and made little or no progress.

Part (b)(ii). Most students who attempted this understood the question to mean finding the

maximum value of N rather than that of the growth rate $\frac{dN}{dt}$, but most had to abandon their attempt. Of those who did differentiate the growth rate expression, most wrote $\frac{d^2N}{dt^2}$, which was condoned; if their attempt led to the maximum growth rate occurring when $N = 250$ they usually completed successfully; those who had worked with inequalities were condoned. Some students thought $T = 250$ was the required answer. Few students saw that the maximum value of the growth rate was at $N = 250$ by inspection, (due to the symmetries of the quadratic function involved) but some concise and correct solutions to the problem were seen.

Q8.

Most candidates demonstrated some understanding of the information given in this in-context question on exponential growth and decay. There were many fully correct solutions.

Most candidates had part (a) correct, with relatively few either not converting 1 hour to 60 minutes or failing to round correctly to the nearest 1000, or not rounding at all.

In part (b)(i), most candidates set up the correct starting-point equation, but they were required to manipulate it to a point where logarithms could be taken before marks were

awarded; many failed to do this. Knowledge of using logarithms varied widely. Many were unable to deal properly with the coefficient and the exponential term and wrote down equations in which t actually should have cancelled out. However, many candidates managed to reach a 'solution' from such incorrect working. In contrast, many other candidates gave a concise derivation of the correct answer.

In part (b)(ii), again many candidates set up the correct starting equation, or something close to it, and some used inequalities, which were accepted. Although many did show clearly how the given equation was derived, others could not handle the exponential terms correctly or made sign errors in manipulating their equation. However, many of those who could not derive the quadratic equation in $e^{\frac{1}{3}t}$ did recognise it as such and solved it successfully.

Pleasingly, most candidates also rejected the $e^{\frac{1}{3}t} = -10$ as impossible. Relatively few tried to find a solution again via taking logarithms immediately, but those who did made no progress and abandoned the question.

Q9.

Most students did part (a)(i) correctly, with very few failing to round to the requested nearest £10. Some confused P and V , so the value of the investment actually went down.

For part (a)(ii), most students showed that they understood the information given and wrote down a correct opening expression, although there was uncertainty over which way the inequality should go. Some then simplified and solved using logarithms efficiently to get the correct answer. Some resorted to trial and improvement, often getting to the correct answer, whilst some students tried a hybrid of both logs and trial and error. Some students failed to give their answer in complete years, thus losing the final mark.

In part (b) also, most students showed they understood the information given and wrote down a correct opening expression. Some made errors in simplifying, but many students who attempted a solution by logarithms, took logarithms incorrectly and their unknown (T or n) cancelled out of their equation. A common error was to combine 1.5 from $\left(\frac{1500}{1000}\right)$ with $\left(1 + \frac{1.5}{100}\right)$ to get a term $n \log 1.5 \left(1 + \frac{1.5}{100}\right)$. Relatively few students got to the correct answer using logarithms. Some realised they had made an error but found the correct answer of 28 from somewhere; it was not always clear what they had done. However, many abandoned their attempt at this point. Many, though, did get a correct answer through trial and improvement, which was accepted for full marks if done correctly, but was worth no marks otherwise.

Q10.

Most candidates scored well on this question although some misinterpreted the information given.

In part (a) most candidates substituted and evaluated correctly. A few, having written the unsimplified form, could not cope with the index and combined the 10 and 2 incorrectly. Some misinterpreted the information and found a mass that decayed to 10 grams in 14 days.

In part (b) most candidates substituted the 16 correctly and then, usually, took log s in

order to

solve for d . Only a minority recognised that $\frac{1}{16} = 2^{-4}$ and so $\frac{d}{8} = 4$. Some candidates began

incorrectly by writing $2^{-\frac{d}{8}} = 16$ which led to a negative answer. Most of these then just dropped the minus sign rather than looking for an error in their working.

In (c) Most candidates set up the correct equation and used logarithms to various bases to solve it. The most common answer here was 53.15, as candidates failed to heed the request to give the answer as an integer. Many also rounded down to 53, misinterpreting what the question was asking.

Not all understood the question and assumed the mass had decayed by 1%, so used 99%.

Q11.

Virtually all candidates had part (a)(i) correct.

In (a)(ii) over 80% of the candidates were correct here, although a variety of methods were seen, ranging from taking the 25th root of 25, to expressions involving logarithms. Those who failed to achieve both marks, usually started correctly, but did not continue to give an expression from which the correct value of k could be found.

In part (b) most candidates were able to use logarithms correctly to solve their equation for t , with few using trial and improvement. However, many then interpreted the result incorrectly rounding down rather than up, not realising that as the result was over 55 it indicated the 56th year and thus 2016. Some candidates used the wrong base year, 1985 being common, and some didn't interpret their answer as a year at all.

Q12.

Most candidates answered part (a)(i) correctly, although a surprising number seemed to think that $A \times 0 = A$.

Most candidates did part (a)(ii) convincingly, usually choosing to evaluate the exponential expression as 0.95 ... and solving for A to demonstrate its value is 60 to two significant figures. Those who did not actually solve for A or did not show A as 59.9...were penalised.

Most candidates started part (b)(i) well, showing they had interpreted the information given correctly. However many could not manipulate their resulting $\ln$ expression into the form required; many, either not knowing what was meant by " a and b are integers" or not taking notice of it, left their result in fractions. Many candidates 'lost' a minus sign during their working whilst some attempted to take the $\ln$ of a negative number.

In part (b)(ii), a few candidates were able to derive the result clearly and efficiently; others managed it after some working that was somewhat difficult to follow whilst others made an error such as dropping a minus sign but still claimed they had obtained the result.

However, many candidates gave up after finding $\frac{dA}{dt}$. A few candidates attempted to

integrate the given differential equation, some successfully recovering the given expression for A , which was both acceptable and impressive.

Although many candidates had part (b)(iii) correct, many errors were also seen, quite a

common one being $4 \times 2 = \frac{1}{2}$. The other common error was to substitute $h = 13$, rather

than $\frac{d\lambda}{dt} = 13$. It was notable that many candidates attempted part (b)(iii) before returning to try and sort out part (b)(ii), which was sensible.

Q13.

Part (a)(i) was answered correctly by virtually all candidates, and attachment the £ sign to the value of A was condoned.

Most candidates earned 2 marks in part (a)(ii), either for giving an answer for k to a greater accuracy than that stated in the question or, more commonly, for isolating k in an unsimplified form. Many variations on a correct expression were seen, but with most candidates giving the simplest; the 36th root of $7000/A$. However, those who did use logarithms usually did it correctly.

In part (b), although most candidates could set up and solve the index equation correctly, the interpretation of the solution showed a common lack of understanding. Although most candidates correctly calculated t to be 56.89, almost all assumed the answer to be 56 and not 57, thus losing the final mark. It seemed that many thought you always round down in this type of question and didn't read the context carefully. Relatively few candidates used a trial and improvement approach, but of those who did many actually demonstrated that the required answer was indeed 57.

Q14.

This question fell into two clear parts: most candidates gained full, or nearly full, marks on part (a), but most were only able to score the first mark in part (b). In part (a)(i), most wrote down the value of A correctly. In part (a)(ii), most candidates obtained an expression for k from which it could be evaluated and so gained the marks. Those who did not give an explicit expression for k lost a mark, and those who just verified that $20k^{60} \approx 2000$ scored no marks. Similarly, candidates were mostly correct in part (a)(iii), but a few made an error in $2008 - 1885$, and some did not round to the nearest 1000.

In part (b), many candidates showed that they had understood the question by setting up the correct equation, but few of those were then able to solve it correctly. The common problem was that, having written down an equation of the form $ak^t = bm^t$, candidates wrote the corresponding log equation as $t \log(ak) = t \log(bm)$ and so, confused by their t apparently cancelling, abandoned their answer. Those who wrote the log equation correctly usually went on to complete the question correctly. The other common mistake was a misunderstanding of the question in which the expression for Q was equated to the result from part (a)(iii); this scored no marks.

Q15.

Many candidates seemed to have little experience of formulating a differential equation as

often a derivative was missing from their answer to part (a)(i). Attempts at a derivative often involved variables other than the given x and t . Some candidates interpreted the question as 'find x in terms of t ', and many expressions involving an exponential function were seen. Some candidates, who essentially had a correct differential equation, also thought they were required to solve it. Few candidates actually responded to the word 'decreasing' in the question, and many had a minus sign missing from an otherwise correct expression.

The given values of 500 and 20000 were often misinterpreted in part (a)(ii); the 500 often being taken to mean time. However, many candidates did indicate they had understood 500 to be a rate and 20000 the current value of x , but often also had a spurious t or another term in x in their differential equation. Some had a correct value of k , but negative, not apparently noting $k > 0$ had been given in the question. Relatively few fully correct answers to part (a) were seen.

Most candidates found the value of A correctly in part (b)(i), the common error being an apparent misinterpretation of the given information and finding A to be 700.

Part (b)(i) of the question was generally done well with many fully correct answers, and very few solutions by a trial and improvement approach. The majority of candidates could manipulate the expression and take logarithms correctly. Some algebraic errors were seen; in particular those who divided 1900 by 2000, rather than subtracting it, gained no further credit. Some first subtracted the 2000, and then tried to take logarithms of the expression as it stood, usually with an error. Some candidates incorrectly used base 10 logarithms. In completing the question most candidates rounded down their result to give the correct year. Some, who had made an error, did not round down and so had their year incorrect. Some candidates misinterpreted the last request in the question giving their answer as the 51st or 52nd year, rather than 2059.

Q16.

The evaluations in part (a) were done correctly by most candidates, although some made numerical errors. In part (a)(ii), the request for three significant figures was largely adhered to, although some interpreted this as three decimal places.

Candidates started part (b) well but many were confused as to how to handle the negative signs.

Many attempts to take logs of negative numbers were seen. Many got to $14 \ln\left(\frac{5}{12}\right)$ or $-14 \ln\left(\frac{5}{12}\right)$ and gave this as their answer. Despite this, the vast majority gave the correct answer of 12 to part (b)(ii), some just ignoring their negative sign or saying it must be positive. They apparently did not respond to a negative answer by reviewing their answer to part (b)(i). Very few candidates explained what they were doing with the negative index to obtain the expected answer. In part (c)(i), very few candidates derived the differential equation with confidence. Some made a start by attempting to differentiate the expression

for x , although, of those, many made the error of bring down the t together with $\frac{1}{14}$. Others attempted to find an expression for t and differentiate that. There was usually little progress beyond these starting points, although several candidates attempted to convince

the examiner that they had found the requested result, with much abuse of the $\ln$ function seen in such attempts.

Some tried to integrate the given differential equation to recover the expression in x , and some very neat solutions were seen this way, although those who did not consider limits or a constant could make no further progress. Again, handling negative signs with logarithmic expressions caused difficulties. Most candidates scored the final mark for

applying the result from part (c)(i), although some did say that $\frac{1}{14}(7) = 2$.

Q17.

This question was answered very well with most candidates gaining nearly full marks. In part (a) many wrote £80 instead of just 80 but this was accepted.

Part (b) was mostly done accurately with many getting to $k = \sqrt[56]{62.5}$ which was accepted as *showing that*. Those who used logarithms lost a mark if they didn't show how $\log 62.5 = 56 \log k$ led to the given value of k .

Virtually all candidates got part (c) correct, with both the approximate answer 200707 and the more exact 200648 being seen, often with several decimal places attached, but this was of no matter as candidates had demonstrated a value greater than £200,000. In part (c)(ii) there were occasional errors although most candidates used logarithms to derive 124.7 correctly whilst a few used trial and improvement to deduce 125. However, many lost the last mark in incorrect interpretation of the year, or through not actually giving a year. The year 2025 was the common wrong answer.

Q18.

Most candidates struggled with part (a) of this question, indicating inexperience, or non understanding, of the formulation of a differential equation from a context situation.

Part (a) Despite the question explicitly requesting a differential equation for $\frac{dx}{dt}$ many candidates just wrote down an expression for x .

A commonly seen differential equation was $\frac{dx}{dt} = kx$ which candidates then proceeded to solve, despite the question stating that a solution of the differential equation was not required. Of those who could translate the given information into correct symbols, a symbol t for time, often found its way into their expression, and then often in an exponential form. Many attempts using the exponential function were seen, suggesting many candidates associating anything to do with differential equations with exponential forms. Those candidates who did give a correct differential equation in part (i) usually found a correct value for the constant of proportionality k . However, there were also many attempts from a partially correct part (i) where the given values of 1000 and 200 were confused, the latter often being taken as a value of time. Other candidates put forward the incorrect argument that the population would be equal to 1200 at time $t = 1$.

Part (b)(i) Virtually all candidates scored at least one mark here; the common error being to round off 5.545 to 5.6 or not to round it at all.

Part (b)(ii) Many candidates solved this equation correctly, although it was not always clear just how they had done it. Most candidates used the technique as per the mark

scheme, although some worked successfully from $e^{30} = \left(\frac{4x}{5000 - x} \right)^4$. Of those who went awry, the error usually involved an incorrect manipulation of the logarithm expression, including ignoring the $\ln$ altogether, or stating $4\ln 4 = \ln 16$. Some started correctly with $7.5 = \ln 4x - \ln (5000 - x)$ but obtained $e^{75} = 4x - (5000 - x)$. Others made relatively simple errors in signs or coefficients in their manipulation towards a solution.

Q19.

Many candidates gained full marks on this question presenting clear, good quality answers.

Part (b)(i) Most candidates knew they were required to separate and integrate and attempted to do so, sometimes with poor notation and incorrect integrals but their intention was clear. Most candidates also included a constant of integration and attempted to find it using the given initial conditions. Some neglected to express t in terms of x with some going straight to the result given in part (b)(ii).

Part (b)(ii) This was often done well but the common error was in taking exponentials, with

$e^{\frac{t}{40}} = e^{\ln 70} + e^{\ln(x-15)}$ being fairly common.



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